Amazon Connect Voice Fundamentals

Course Summary

In this course, you learned about Amazon Connect telephony and voice channels.

Benefits of the Amazon Connect telephony

Reliability and redundancy

Amazon Connect is a highly reliable and scalable cloud telephony solution for mission-critical contact center operations. It provides resilient network connections to multiple telephony carriers over the PSTN, with automatic rerouting over redundant paths if any connection fails. In addition to traditional PSTN calling, Amazon Connect supports internet in-app calling, web calling, and video calling options. The telephony layer runs on fully-managed SBCs spread across multiple Availability Zones, ensuring service continuity even if an entire AZ goes down within a region.

For even higher levels of redundancy and business continuity, Amazon Connect offers multi-region redundancy capabilities. The Global Resiliency feature allows inbound telephony to operate in an active-active mode across two AWS Regions in the US and Europe. This provides an extra layer of reliability by enabling seamless failover between regions in the event of a regional outage or disaster. Overall, Amazon Connect is engineered to deliver highly durable and resilient cloud contact center services.

Global reach

Amazon Connect is built on AWS's global infrastructure. This allows for high-quality voice communications with low latency anywhere in the world. Amazon Connect has partnered with telephony carriers globally, enabling businesses to provision local phone numbers in the countries where their customers are based. This includes the ability to use universal international freephone numbers (UIFNs) at no cost, providing a consistent global experience.

In addition to traditional phone calling, Amazon Connect offers multiple channels for customers to reach contact centers, including in-app, web, and video calling options that connect directly over the internet. Between local phone numbers, toll-free numbers, UIFNs, and internet-based calling, Amazon Connect provides organizations with flexible options to engage with their customers regardless of their location. This omnichannel approach powered by AWS's global footprint allows for seamless customer communications worldwide.

Fully managed telephony

Amazon Connect provides a fully-managed telephony infrastructure that handles all the complex back-end requirements involved in running a contact center operation. This includes managing relationships with telecommunications carriers, maintaining and patching telephony equipment, ensuring compliance with regulations in different countries, and securing the overall environment.

By offloading this telephony undifferentiated heavy lifting to AWS, organizations can focus their time and resources on their core business priorities rather than the cumbersome tasks of managing contact center infrastructure themselves. Amazon Connect allows companies to innovate and improve their customer experience, agent performance, and operational processes without being burdened by the complexities of telephony infrastructure management. The service enables businesses of all sizes to run an enterprise-grade contact center by tapping into the scalability and reliability of AWS's global cloud infrastructure.

Economies of scale

Amazon Connect offers a flexible and cost-effective pricing model for organizations looking to migrate their contact center operations to the cloud. Instead of requiring large upfront investments, Amazon Connect utilizes a pay-as-you-go approach with no upfront fees. This allows companies to avoid overprovisioning capacity and only pay for the actual usage, rather than maintaining their own telephony infrastructure.

One of the key benefits of Amazon Connect is that as its global usage of telephony services increases, AWS is able to negotiate better deals with telephony carriers due to the high call volumes. These savings get passed along to all Connect customers in the form of lower prices. This economy of scale means that as more organizations adopt Connect, the more affordable it becomes for everyone using the service thanks to AWS's negotiating power with carriers. Overall, Amazon Connect provides a low-risk, low-cost entry point for cloud contact center capabilities.

Amazon Connect Telecoms Country Coverage Guide

The Amazon Connect Telecoms Country Coverage Guide provides information on the telecom services offered by Amazon Connect in various countries through partnerships with carrier providers. It details the service types, number availability, outbound calling capabilities, number porting options, multi-carrier support, and more.

The guide aims to help organizations assess the telecom features available in different countries to better serve their customers while adhering to relevant regulations.

Claiming and Porting phone numbers

Amazon Connect offers a pool of phone numbers that you can instantly claim and use with your contact center. These numbers are available for fast setup and are owned and managed by Amazon Connect. You can use them while you keep your Amazon Connect instance. Amazon Connect users can be granted security profile permissions to claim phone numbers in the console.

Porting Phone Numbers to Amazon Connect: Number porting is the process of moving a telephone number from one communications provider to another. When migrating your contact center workloads to Amazon Connect, you have an option to claim a new number or port your existing phone numbers. While claiming numbers is faster, porting allows you to keep your existing numbers and offer a seamless uninterrupted services transition for your customers.

Verifying Porting Availability and Initiating the Process: You can use the Amazon Connect Telecoms Country Coverage Guide to verify if number porting is supported for the region and country where your contact center operates. Once you confirm porting availability, you can open an AWS Support ticket to initiate the number porting process. Follow the steps described in the Amazon Connect Administrator Guide.

After Porting Completes: Once the porting process is complete and successful porting is verified, the ported numbers will appear in your Amazon Connect console, and you can associate them with Amazon Connect Flows. It is best practice to test the ported numbers by placing test calls to ensure the associated contact Flow is successfully activated, ensuring a seamless transition for your customers.

Benefits of in-app, web, and video calling

Cost effective customer access: Users can initiate voice calls for any inquiries or assistance needed. They can contact customer service without incurring expensive international calling charges or roaming costs.

Convenience: Using the same internet-based calling Amazon Connect capabilities, users can initiate voice calls to agents directly from their website.

Faster resolution: When users need to provide a better visual detail of their issue, they can activate real-time video sharing within the same communication.

Reduced customer effort: When users initiate calls through in-app, web, or video calling, contact centers can perform authentication based on the information available in the mobile app or website. The availability of this data reduces customer effort, and streamlines the authentication process by eliminating the need for additional verification. User's language preference or location can be used to route the call to an agent skilled to communicate in the user's preferred language.

Added service redundancy and resiliency: In the event of a telephony carrier outage or disruption of PSTN carrier services, users can use in-app, web, and video calling. The additional built-in redundancy helps organizations provide better service and business continuity for their customers.

Pricing structure

Amazon Connect offers a pay-as-you-go pricing model with no long-term commitments or minimum monthly fees. Organizations are charged based on the number of minutes they use Amazon Connect to engage with customers. The per-minute rates vary depending on the country, services used, and types of calls.

There are three distinct charges for Amazon Connect's public switched telephone network (PSTN) calling: voice service usage, telecommunication service usage, and number usage. Voice service usage covers the cost of Amazon's software that handles call routing and management. Telecommunication service usage covers the cost of connectivity between Amazon and telecom providers for inbound and outbound calls. Number usage covers any phone numbers, such as DIDs or TFNs, provisioned through Amazon Connect.

Amazon Connect charges for internet-based calling are divided into three categories: Amazon Connect service usage, audio service usage, and video service usage. For the Amazon Connect service, charges are billed per second with a minimum duration of 10 seconds.

The audio service charges are also billed per second from the time the contact attempt is initiated, with a minimum of 10 seconds. The audio usage is determined by the total seconds the customer is connected to Amazon Connect. Video service usage is charged by the second as well, with a 10 second minimum. Video charges apply separately for each party that has video activated, for the duration their video is turned on.

Considerations

Security

Amazon Connect follows a shared responsibility security model where AWS manages security of the cloud infrastructure and services, while customers are responsible for securing their data and applications. All data in Amazon Connect is encrypted in transit and at rest. AWS handles encryption, access control, and security at the infrastructure and platform layers. Customers control security aspects related to their specific use case like network access, user permissions, and protecting their customer data.

For PSTN calls routed over the public internet, signaling data is encrypted with TLS and the audio media is encrypted with SRTP. For internet-based calls to an agent's browser, the WebSocket connection is encrypted with TLS and the audio media uses DTLS-SRTP encryption. Amazon Connect's architecture and protocols ensure end-to-end encryption and protection of voice communications.

Resiliency and redundancy
Carrier redundancy involves claiming phone numbers across multiple telephony carriers in countries where Amazon Connect partners with multiple carriers. This provides continuous service availability and ensures if one carrier experiences an outage, numbers hosted on other carriers remain reachable. The Amazon Connect Telecoms Country Coverage Guide indicates if the multi-carrier feature is available in the country. In the U.S., Amazon Connect automatically provides redundant toll-free numbers routed across multiple carriers.

High call volumes refer to large numbers of concurrent calls arriving simultaneously. By using phone numbers from multiple carriers, the concurrent call load can be distributed across different carriers. This increases the overall concurrency capacity of the contact center.

Call quality
To ensure high-quality voice interactions on Amazon Connect, organizations must consider several key factors. Network and agent workstation requirements are crucial, with Amazon Connect offering options for network connectivity setup and a minimum of 2GB dedicated memory for agent workstations. Virtual Desktop Infrastructure (VDI) deployments add complexity, requiring optimal audio processing on local devices and specific setup guidelines. Network latency, the time taken for data packets to travel, should be kept below 500ms end-to-end for good call audio quality, which can be verified using Amazon Connect's Endpoint Test Utility.

Telephony latency, introduced when calls are forwarded from carriers to Amazon Connect or across distant locations, can affect audio quality. Monitoring agent workstations and considering partner solutions for call quality monitoring is recommended. Amazon Connect provides quality metrics in contact records, and organizations should configure CloudWatch alerts for relevant metrics. Calls transferred out of and back into Amazon Connect can introduce additional Public Switched Telephone Network (PSTN) latency, further impacting the audio quality.